

一. 1. $1-(1-p)^n$ 2. $21/4$ 3.D 4. $\sqrt{2}$ 5. $\chi^2(2)$ 6. $H_0: \mu \leq a, \alpha = 0.05$ 7.

$$(\bar{X} - \frac{S}{\sqrt{n}}t_{a/2}(n-1), \bar{X} - \frac{S}{\sqrt{n}}t_{a/2}(n-1)) \quad 8. \quad \frac{1}{(t_1+t_2)a}(1-e^{-a(t_1+t_2)})$$

二. 0.22

三.解:

$$f_X(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x < 0 \end{cases} \quad \text{与} \quad f_Y(y) = \begin{cases} e^{-y} & y > 0 \\ 0 & y < 0 \end{cases}.$$

X, Y 相互独立

$$\begin{aligned} \text{当 } z > 0 \text{ 时,} \quad F_Z(z) &= P\{\max(X, Y) \leq z\} = P\{X \leq z, Y \leq z\} \\ &= P\{X \leq z\}P\{Y \leq z\} = F_X(z)F_Y(z) \end{aligned}$$

$$f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z) = 2e^{-2z}(1-e^{-z}) + (1-e^{-2z})e^{-z} = e^{-z} + 2e^{-2z} - 3e^{-3z}$$

$$\text{故} \quad f_Z(z) = \begin{cases} e^{-z} + 2e^{-2z} - 3e^{-3z} & z > 0 \\ 0 & \text{其他} \end{cases}$$

四. $-260 \cdot 1/5 + 150 \cdot 4/5 = 68$

$$\text{五. } P(X=k) = (1-p)^{k-1}p, \quad p = \frac{1}{X}$$

$$\text{六. 设 } X_i = \begin{cases} 1 & \text{第 } i \text{ 台彩电为次品且未被查出} \\ 0 & \text{其他} \end{cases} \quad i = 1 \sim 2 \times 10^5$$

$$E(X_i) = 5 \times 10^{-6}, \quad D(X_i) = 5 \times 10^{-6}(1 - 5 \times 10^{-6})$$

$$\text{经检验后的次品数 } Y = \sum_{i=1}^{2 \times 10^5} X_i, \quad E(Y) = 1, \quad D(Y) = 1 - 5 \times 10^{-6},$$

由中心极限定理, 近似地有 $Y \sim N(1, 1 - 5 \times 10^{-6})$

$$P(Y > 3) = 1 - P(Y \leq 3) \approx 1 - \Phi\left(\frac{3-1}{\sqrt{1-5 \times 10^{-6}}}\right) \approx 1 - \Phi(2) = 0.0228.$$